Chapter 34: Chemical Coordination: Endocrine System

Introduction to Endocrinology

- **Endocrinology** is the study of chemical coordination in animals.
- **Hormones** are chemical messengers synthesized by specialized cells, released into the blood in small amounts, and transported to distant **target cells** where they initiate physiological responses.
- **Endocrine glands** are ductless glands that secrete hormones directly into the blood.
- **Exocrine glands** have ducts and secrete products (e.g., sweat, saliva, digestive enzymes) onto a free surface.
- **Neurosecretory cells** are specialized nerve cells that release **neurohormones** into the blood.
- Historical milestones:
  - **Bayliss and Starling** (1902) discovered **secretin**, the first hormone, demonstrating chemical control of pancreatic secretion. Starling coined the term "hormone".
  - **Berthold** (1849) demonstrated that testes secrete a blood-borne signal controlling male characteristics in roosters.

Figure 1: Founders of endocrinology. A, Sir William H. Bayliss (1860 to 1924). B, Ernest H. Starling (1866 to 1927).

Mechanisms of Hormone Action

- Hormones affect only target cells that have specific **receptors**.
- **Membrane-Bound Receptors**:
  - Used by peptide hormones, epinephrine, and other molecules too large or polar to cross the cell membrane.
  - Hormone acts as a **first messenger**.

- Binding triggers a **second-messenger** system (e.g., **cyclic AMP (cAMP)**, cGMP, Ca^{2+}, IP3, DAG) inside the cell.
- Signal amplification occurs via enzyme cascades (e.g., kinases).

- **Nuclear Receptors**:
  - Used by **steroid hormones** and **thyroid hormones** (lipid-soluble).
  - Hormones diffuse into the cell and bind to cytoplasmic or nuclear receptors.
  - The hormone-receptor complex acts as a **gene regulatory protein**, altering gene transcription (DNA to mRNA) and protein synthesis.

Figure 2: Mechanisms of hormone action. Peptide hormones and epinephrine act through second-messenger systems, as for example, cyclic AMP, shown here. The combination of hormone with a membrane receptor stimulates the enzyme adenylate cyclase to catalyze formation of cyclic AMP (second messenger). Thyroid hormones bind to a membrane receptor and are transported into the cell by active transport. There they bind to cytoplasmic receptors that are transported into the nucleus to alter gene transcription. Steroid hormones penetrate the cell membrane to combine with cytoplasmic or nuclear receptors that alter gene transcription.

Control of Secretion Rates

- **Negative Feedback**:
  - The most common control mechanism.
  - Output of the system counteracts the initial stimulus, maintaining a set point.
  - Example: CRH -> ACTH -> Cortisol. High cortisol inhibits CRH and ACTH release.
- **Positive Feedback**:
  - Output reinforces the stimulus, leading to an explosive event (e.g., oxytocin in childbirth). Requires a shutoff mechanism.

Figure 3: Negative feedback systems. A specific example shows the control of cortisol secretion: CRH from the hypothalamus stimulates ACTH from the pituitary, which stimulates cortisol from the adrenal cortex. Cortisol inhibits CRH and ACTH release. A general diagram shows how hormone levels feed back to endocrine cells to regulate secretion.

Invertebrate Hormones

- Invertebrates use peptides, steroids, and terpenoids.
- **Molting and Metamorphosis in Insects**:
  - Controlled by interaction of two hormones:
    - **Ecdysone** (molting hormone): A steroid produced by the **prothoracic gland**. Promotes molting and adult differentiation.
    - **Juvenile Hormone (JH)**: A terpenoid produced by the **corpora allata**. Favors retention of juvenile characteristics.
  - **Prothoracicotropic Hormone (PTTH)**: A neurohormone from the brain that stimulates ecdysone secretion.
  - High JH + Ecdysone -> Larval-to-larval molt.
  - Low JH + Ecdysone -> Larval-to-pupal molt -> Metamorphosis.

Figure 4: Endocrine control of molting in a moth, typical of insects having complete metamorphosis. Juvenile hormone and ecdysone interact to control molting and pupation. PTTH from the brain stimulates the prothoracic gland to secrete ecdysone, which causes molting. Juvenile hormone from the corpora allata determines the nature of the molt (larval-larval vs. larval-pupal-adult). Chromosome puffing indicates gene activation during these stages.

Vertebrate Endocrine Glands and Hormones

Hypothalamus and Pituitary Gland

- The **Pituitary Gland** (Hypophysis) has two parts:
  - **Anterior Pituitary** (Adenohypophysis): Derived from the roof of the mouth. Connected to the hypothalamus by a **portal circulatory system**.
  - **Posterior Pituitary** (Neurohypophysis): Derived from the brain. Connected to the hypothalamus by axons of neurosecretory cells.

Figure 5: Human hypothalamus and pituitary gland. The posterior lobe is connected directly to the hypothalamus by axons of neurosecretory cells. The anterior lobe is indirectly connected to the hypothalamus by a portal circulation (shown in red) beginning in the base of the hypothalamus and ending in the anterior pituitary.

- **Hypothalamic Control**:
  - The hypothalamus secretes **releasing hormones** and **inhibiting hormones** that regulate the anterior pituitary via the portal system.

Figure 6: Relationship of hypothalamic, pituitary, and target-gland hormones. The hormone sequence controlling the release of cortisol from the adrenal cortex is used as an example. Hypothalamic CRH travels via the portal system to the anterior pituitary, stimulating ACTH release, which enters the systemic circulation to stimulate the adrenal cortex.

- **Radioimmunoassay (RIA)**:
  - Developed by **Yalow and Berson** (1960).

- A sensitive technique to measure hormone levels using antibodies and radioisotopes.

- **Anterior Pituitary Hormones**:
  - **Tropic Hormones** (regulate other glands):
    - **Thyroid-Stimulating Hormone (TSH)**: Stimulates thyroid.
    - **Adrenocorticotropic Hormone (ACTH)**: Stimulates adrenal cortex.
    - **Follicle-Stimulating Hormone (FSH)**: Gonadotropin (gamete production).
    - **Luteinizing Hormone (LH)**: Gonadotropin (sex steroids, ovulation).
  - **Non-Tropic Hormones**:
    - **Prolactin (PRL)**: Milk production, parental behavior, water balance.
    - **Growth Hormone (GH)**: Body growth, metabolism, IGF production.
    - **Melanocyte-Stimulating Hormone (MSH)**: Pigment dispersion in ectotherms.

Figure 7: Hormones of the Vertebrate Pituitary. Lists hormones of the Adenohypophysis (Anterior lobe), Intermediate lobe, and Neurohypophysis (Posterior lobe), their chemical nature, principal actions, and hypothalamic controls.

- **Posterior Pituitary Hormones**:
  - Stored and released by the posterior pituitary but synthesized in the hypothalamus.
  - **Oxytocin**: Uterine contractions (birth), milk ejection.
  - **Vasopressin** (ADH): Water reabsorption in kidneys, vasoconstriction.
  - **Vasotocin**: Ancestral hormone in non-mammals (water balance).

Figure 8: Posterior pituitary hormones of mammals. Both oxytocin and vasopressin consist of eight amino acids (octapeptides). They are identical except for amino-acid substitutions in two positions (Ile/Phe and Leu/Arg).

Pineal Gland

- Derived from the diencephalon.
- In ectotherms (lampreys, some amphibians, lizards), it contains photoreceptors and is often called a **"third eye"**.
- In birds and mammals, it is purely glandular and secretes **melatonin**.
- Regulates **circadian rhythms** (biological clock) and seasonal reproduction (photoperiodism).
- Secretion is high at night and low during the day.

Other Chemical Messengers

- **Brain Neuropeptides**: Endorphins and enkephalins (pain modulation, pleasure).
- **Prostaglandins**: Local tissue factors derived from fatty acids (inflammation, smooth muscle contraction).
- **Cytokines**: Immune system messengers.

Hormones of Metabolism

Thyroid Gland

- Follicles accumulate iodine to synthesize **Triiodothyronine (T3)** and **Thyroxine (T4)**.
- Functions:
  - Regulate metabolic rate and heat production (uncoupling oxidative phosphorylation).
  - Essential for normal growth and nervous system development.

Figure 9: Appearance of thyroid gland follicles viewed through the microscope. When inactive, follicles are distended with colloid (stored T3 and T4) and epithelial cells are flattened. When active, the colloid disappears as hormones are secreted, and epithelial cells become enlarged.

- Control **metamorphosis** in amphibians (e.g., frogs).
- Transformation from aquatic larva to terrestrial adult is triggered by thyroid hormones.

Figure 10: Effect of thyroid hormones (T3 and T4) on growth and metamorphosis of a frog. The release of TRH from the hypothalamus at the end of premetamorphosis sets in motion hormonal changes (increased TSH, T3 and T4) leading to metamorphosis. Thyroid hormone levels are maximal at the time forelimbs emerge.

- **Goiter**: Enlargement of the thyroid gland, often due to iodine deficiency (high TSH stimulation).

Figure 11: A large goiter caused by iodine deficiency. Overstimulation by excess TSH causes the gland to enlarge enormously, as the thyroid gland is stimulated to extract enough iodine from the blood to synthesize the body's requirement for thyroid hormones.

Calcium Metabolism

- Regulated by three hormones:
  - **Parathyroid Hormone (PTH)**: From parathyroid glands. Increases blood Ca^{2+} (bone resorption, kidney reabsorption, Vitamin D activation).
  - **Calcitonin**: From thyroid C cells. Decreases blood Ca^{2+} (inhibits bone resorption).

Figure 12: How rate of secretion of parathyroid hormone (PTH) and calcitonin respond to changes in blood-calcium level in a mammal. PTH secretion rises as plasma calcium falls, while calcitonin secretion rises as plasma calcium rises.

- **1,25-Dihydroxyvitamin D3**: Activated form of Vitamin D. Increases intestinal Ca^{2+} absorption.

• Vitamin D deficiency causes **rickets** (weak bones).

Figure 13: Regulation of blood calcium in birds and mammals. PTH stimulates osteoclasts to release calcium from bone, increases calcium reabsorption in kidneys, and stimulates formation of 1,25-dihydroxyvitamin D3 (which aids intestinal absorption). Calcitonin stimulates osteoblasts to deposit calcium in bone and allows calcium excretion.

Adrenal Glands

- **Adrenal Medulla** (inner): Secretes catecholamines.
  - **Epinephrine** (Adrenaline) and **Norepinephrine**.
  - “Fight or flight” response (sympathetic nervous system).

Figure 14: Paired adrenal glands of humans, showing gross structure and position on the upper poles of the kidneys. Steroid hormones are produced by the cortex. The sympathetic hormones epinephrine and norepinephrine are produced by the medulla.

- **Adrenal Cortex** (outer): Secretes steroid hormones.

- **Glucocorticoids** (e.g., **Cortisol**): Regulate glucose metabolism (gluconeogenesis), stress response, immune suppression.
- **Mineralocorticoids** (e.g., **Aldosterone**): Regulate salt and water balance (Na^+ reabsorption, K^+ excretion).
- **Androgens**: Sex steroids.

Figure 15: Hormones of the adrenal cortex. Cortisol (a glucocorticoid) and aldosterone (a mineralocorticoid) are two of several steroid hormones synthesized from cholesterol in the adrenal cortex.

Pancreas

- **Islets of Langerhans** (endocrine portion):
  - **Alpha cells**: Secrete **Glucagon** (raises blood glucose).
  - **Beta cells**: Secrete **Insulin** (lowers blood glucose).

Figure 16: The pancreas is composed of two kinds of glandular tissue: exocrine acinar cells that secrete digestive juices, and endocrine islets of Langerhans. The islets secrete insulin (beta cells), glucagon (alpha cells), and pancreatic polypeptide directly into the blood.

- **Diabetes Mellitus**:
  - Type 1: Insulin deficiency (autoimmune destruction of beta cells).
  - Type 2: Insulin insensitivity (often linked to obesity).
- **Discovery of Insulin**:
  - **Banting and Best** (1921) successfully extracted insulin and treated diabetic dogs.

Figure 17: Charles H. Best and Sir Frederick Banting in 1921 with the first dog to be kept alive by insulin.

Adipose Tissue

- White adipose tissue is an endocrine organ secreting **adipokines**.
- **Leptin**: Regulates body weight and satiety (signals fat stores to the brain).

- **Adiponectin**: Increases insulin sensitivity.